

Lab 2 Submission

Instance Tracking

First Instance

- instance-id: i-0d11ca8c33861f61f
- availability-zone: ap-southeast-1b

Second Instance

- instance-id: i-018d9aaa0a1889e5b
- availability-zone: ap-southeast-1a

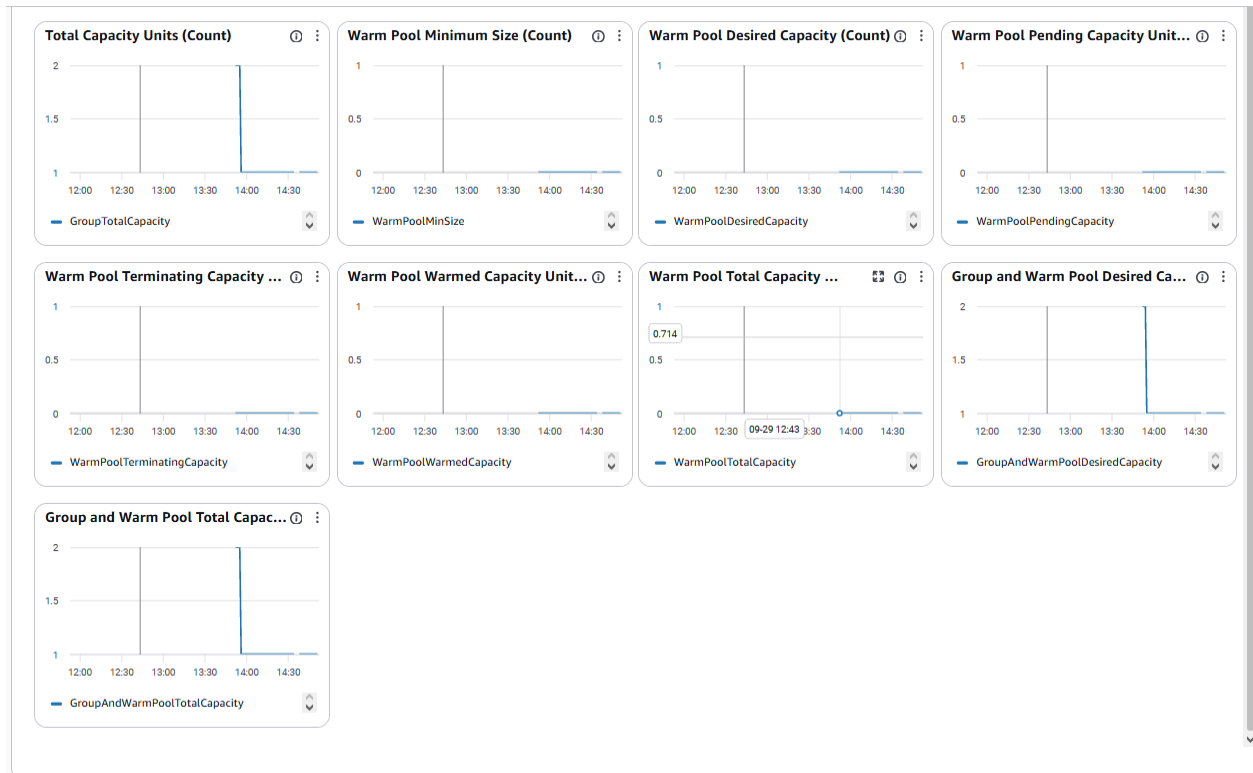
Proof (Screenshots)

1. Activity History: Add a screenshot showing the Auto Scaling group's Activity history when the second instance launched.

The screenshot displays the AWS Management Console interface for the Auto Scaling group 'ccsad-g01-asg'. The left sidebar shows the navigation menu with categories: Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The main content area shows the 'Activity history (3)' for this group. The history table lists three events:

Status	Description	Cause	Start time
Successful	Terminating EC2 instance: i-0d11ca8c33861f61f	At 2026-09-29T13:54:56Z a monitor alarm TargetTracking-ccsad-g01-asg-AlarmLow-2e0feb5f-af58-45b9-9852-64cab70133f in state ALARM triggered policy Target Tracking Policy changing the desired capacity from 2 to 1. At 2026-09-29T13:55:08Z an instance was taken out of service in response to a difference between desired and actual capacity, shrinking the capacity from 2 to 1. At 2026-09-29T13:55:09Z instance i-0d11ca8c33861f61f was selected for termination.	2026 September 29, 09:55:09 PM +08:00
Successful	Launching a new EC2 instance: i-018d9aaa0a1889e5b	At 2026-09-29T13:34:35Z a monitor alarm TargetTracking-ccsad-g01-asg-AlarmHigh-81af775d-37c4-4f4e-8b08-e32c5876ea7d in state ALARM triggered policy Target Tracking Policy changing the desired capacity from 1 to 2. At 2026-09-29T13:34:47Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 1 to 2.	2026 September 29, 09:34:49 PM +08:00
Successful	Launching a new EC2 instance: i-0d11ca8c33861f61f	At 2026-09-29T13:20:19Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 1. At 2026-09-29T13:20:23Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 1.	2026 September 29, 09:20:24 PM +08:00

2. CloudWatch Alarm: Add a screenshot of the target tracking alarm in the "In alarm" state.



Questions

1. Why did the group stop at 2 instances?
The Auto Scaling Group was configured with a maximum capacity of 2 instances. When CPU utilization increased above the 30% target, Auto Scaling launched another instance, increasing the group from 1 to 2. It could not automatically launch a third instance because the maximum capacity was set to 2.
2. Why did terminating an instance by hand not remove the cost?
Terminating an instance removes that particular EC2 instance, but the Auto Scaling Group tries to maintain its configured desired capacity. Since the desired

capacity was 2, the group automatically launched a replacement instance after one was terminated. Therefore, the replacement instance could continue generating EC2 charges.

3. Why is the target value set to your assigned value (e.g., 30-85 percent) instead of 99 percent?

The target value tells Auto Scaling when to add capacity. A target of 99% would allow CPU utilization to become extremely high before scaling out. This could cause slow response times or poor performance while the new instance is being launched. Our assigned target of 30% gives Auto Scaling more room to respond before CPU usage becomes excessive.

4. What did the automatic cutoff protect us from?

The automatic cutoff protected us from leaving the lab's Auto Scaling resources running indefinitely and potentially accumulating additional AWS charges. It also prevented the CPU-burning workload from continuing unnecessarily after the lab period.

5. What changes when a load balancer sits in front of the group?

With a load balancer, users normally access the load balancer instead of individual instance public IP addresses. The load balancer distributes incoming requests among healthy instances in the Auto Scaling Group. When Auto Scaling launches or replaces an instance, the load balancer can add the healthy instance to the pool and stop sending traffic to unhealthy instances. This makes scaling more transparent to users because they continue using one endpoint instead of manually switching between instance IP addresses.

Team Contribution Log

Lab Section	Driver	Navigator	Recorder	Reviewer
Step 1 & 2	Benitez	Calica	Banhao	Camacho
Step 3 & 4	Camacho	Banhao	Calica	Benitez

Step 5 to 7	Banhao	Camacho	Benitez	Calica
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